

Puppet Mastery Part-6: Practical 15 (Flagship Standalone Practical-4) - Enterprise Puppet Automation: Advanced Custom Functions, Hiera/APL, Facts, EPP Templates, Dynamic Resources, and Multi-Layer Configuration Management

Master:

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments
ubuntu@puppet-master:/etc/puppetlabs/code/environments$ find .
.
./development
./development/manifests
./development/manifests/site.pp
./development/modules
./development/modules/dynamicinfra
./development/modules/dynamicinfra/templates
./development/modules/dynamicinfra/manifests
./development/modules/dynamicinfra/functions
./production
./production/data
./production/data/nodes
./production/data/nodes/puppet-agent01.yaml
./production/data/production.yaml
./production/hiera.yaml
./production/manifests
./production/manifests/site.pp
./production/modules
./production/modules/dynamicinfra
./production/modules/dynamicinfra/templates
./production/modules/dynamicinfra/templates/deployment-report.pp
./production/modules/dynamicinfra/templates/system-info.epp
./production/modules/dynamicinfra/templates/deployment-report.epp
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments
./production/data/nodes/puppet-agent01.yaml
./production/data/production.yaml
./production/hiera.yaml
./production/manifests
./production/manifests/site.pp
./production/modules
./production/modules/dynamicinfra
./production/modules/dynamicinfra/templates
./production/modules/dynamicinfra/templates/deployment-report.pp
./production/modules/dynamicinfra/templates/system-info.epp
./production/modules/dynamicinfra/templates/deployment-report.epp
./production/modules/dynamicinfra/manifests
./production/modules/dynamicinfra/manifests/install.pp
./production/modules/dynamicinfra/manifests/config.pp
./production/modules/dynamicinfra/manifests/service.pp
./production/modules/dynamicinfra/manifests/init.pp
./production/modules/dynamicinfra/manifests/report.pp
./production/modules/dynamicinfra/functions
./production/modules/dynamicinfra/functions/normalize_role.pp
./production/modules/dynamicinfra/functions/deployment_identity.pp
./production/modules/dynamicinfra/functions/environment_label.pp
./production/modules/dynamicinfra/functions/package_manager.pp
./production/environment.conf
ubuntu@puppet-master: /etc/puppetlabs/code/environments$
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Aug 10 18:22:25 2026 from 152.58.79.36
ubuntu@puppet-master:~$ cd /etc/puppetlabs/code/environments/production/manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ sudo cat site.pp
node 'puppet-agent01'{
  include dynamicinfra
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd modules/dynamicinfra/manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynamicinfra/manifests$ ls
config.pp  init.pp  install.pp  report.pp  service.pp
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
cinfra/manifests$ cat init.pp
class dynamicinfra(
  String $server_role,
  String $environment_name,
  String $output_directory,
  Array[String] $packages
){
  class{'dynamicinfra::install':
    packages=>$packages
  }

  class{'dynamicinfra::config':
    server_role=>$server_role,
    environment_name=>$environment_name,
    output_directory=>$output_directory
  }

  class{'dynamicinfra::service': }

  class{'dynamicinfra::report':
    server_role=>$server_role,
    environment_name=>$environment_name,
    output_directory=>$output_directory
  }
}

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
}
Class['dynamicinfra::install']->Class['dynamicinfra::config']
->Class['dynamicinfra::service']->Class['dynamicinfra::report']
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ cat config.pp
class dynamicinfra::config(
  String $server_role,
  String $environment_name,
  String $output_directory
){
  $normalized_role=dynamicinfra::normalize_role($server_role)
  $environment_label=dynamicinfra::environment_label($environment_name)
  $deployment_identity=dynamicinfra::deployment_identity($server_role,$environmen
t_name)

  $os_family=$facts['os']['family']
  $package_manager=dynamicinfra::package_manager($os_family)
  $os_name=$facts['os']['name']
  $hostname=$facts['networking']['hostname']
  $ip=$facts['networking']['ip']

  $service_name=$facts['os']['family']?{
    'debian'=>'apache2',
    'redhat'=>'httpd',

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
}
'default'=>'apache2'
}

file{$output_directory:
  ensure=>directory,
  owner=>'root',
  group=>'root',
  mode=>'0755'
}

file{"${output_directory}/system-info.txt":
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',
  content=>epp(
    'dynamicinfra/system-info.epp',
    {
      'server_role'=>$server_role,
      'normalized_role'=>$normalized_role,
      'environment_label'=>$environment_label,
      'deployment_identity'=>$deployment_identity,
      'environment_name'=>$environment_name,
      'os_family'=>$os_family,
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
      'os_name'=>$os_name,
      'hostname'=>$hostname,
      'ip'=>$ip,
      'package_manager'=>$package_manager
    }
  ),
  notify=>Service[$service_name]
}
}

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynami
infra/manifests$ cat install.pp
class dynamicinfra::install(
  Array[String] $packages,
){
  $web_package=$facts['os']['family']?{
    'Debian'=>'apache2',
    'RedHat'=>'httpd',
    default=>'apache2'
  }

  package{$packages:
    ensure=>installed
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
package{$web_package:
  ensure=>installed
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ cat service.pp
class dynamicinfra::service {
  $service_name = $facts['os']['family']?{
    'Debian' => 'apache2',
    'RedHat' => 'httpd',
    default => 'apache2'
  }
  service{$service_name:
    ensure=>running,
    enable=>true
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ cat reports.pp
cat: reports.pp: No such file or directory
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ cat report.pp
class dynamicinfra::report(
  String $server_role,

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
cinfra/manifests$ cat report.pp
class dynamicinfra::report(
  String $server_role,
  String $environment_name,
  String $output_directory
){
  file{"${output_directory}/deployment-report.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>epp(
      'dynamicinfra/deployment-report.epp',
      {
        'server_role'=>$server_role,
        'environment_name'=>$environment_name,
        'hostname'=>$facts['networking']['hostname'],
        'ip'=>$facts['networking']['ip'],
        'os'=>$facts['os']['name']
      }
    )
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/templates$ cat deployment-report.epp
=====
DEPLOYMENT REPORT
=====

Server Role      : <%= $server_role %>
Environment      : <%= $environment_name %>

Hostname         : <%= $hostname %>
Primary IP       : <%= $ip %>
Operating System : <%= $os %>

Status           : Managed by Vishnu
=====
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/templates$ cat system-info.epp
=====
ENTERPRISE INFRASTRUCTURE REPORT
=====

Server Role      : <%= $server_role %>
Normalized Role  : <%= $normalized_role %>
Environment      : <%= $environment_name %>
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
=====

Server Role      : <%= $server_role %>
Normalized Role  : <%= $normalized_role %>
Environment      : <%= $environment_name %>
Environment Label : <%= $environment_label %>
Deployment Identity: <%= $deployment_identity %>

Hostname         : <%= $hostname %>
Primary IP       : <%= $ip %>

Operating System : <%= $os_name %>
OS Family        : <%= $os_family %>

Package Manager  : <%= $package_manager['name'] %>
Package Command  : <%= $package_manager['command'] %>

Generated By     : Vishnu
=====
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/templates$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd ..
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
GNU nano 6.2 production.yaml
dynamicinfra::environment_name: "Production"
dynamicinfra::output_directory: "/opt/company"
dynamicinfra::packages:
- tree
- curl
- htop

[ Read 6 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
GNU nano 6.2 nodes/puppet-agent01.yaml
dynamicinfra::server_role: "Web Server"

[ Read 1 line ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

Functions:

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mod...  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami  
cinfra/functions$ ls  
deployment_identity.pp  normalize_role.pp  
environment_label.pp    package_manager.pp  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami  
cinfra/functions$ sudo cat deployment_identity.pp  
function dynamicinfra::deployment_identity(  
  String $role,  
  String $environment  
)>>String{  
  "${dynamicinfra::environment_label($environment)} -  
    ${dynamicinfra::normalize_role($role)}"  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami  
cinfra/functions$ cat normalize_role.pp  
function dynamicinfra::normalize_role(String $role='Gen Server') >> String{  
  case $role{  
    'Web Server': {'web-server'}  
    'Application Server': {'application-server'}  
    'Database Server': {'database-server'}  
    'Gen Server': {'generic-server'}  
  
    default: {  
      fail("Unsupported server role '${role}'")  
    }  
  }  
}
```

```
fail("Unsupported server role '${role}'")  
}  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami  
cinfra/functions$ cat environment_label.pp  
function dynamicinfra::environment_label(String $environment)>>String{  
  case $environment {  
    'Production': {'PRODUCTION ENVIRONMENT'}  
    'Development': {'DEVELOPMENT ENVIRONMENT'}  
    default: {"UNKNOWN ENVIRONMENT: ${environment}"}  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami  
cinfra/functions$ cat package_manager.pp  
function dynamicinfra::package_manager(  
  String $os_family='Debian') >> Hash[String,String] {  
  
  case $os_family{  
    'Debian' : { {'name' => 'APT', 'command'=>'apt' } }  
    'Redhat' : { {'name' => 'DNF/YUM', 'command'=>'dnf' } }  
    default : { fail("Unsupported OS family '${os_family}'") }  
  }  
}
```


Agent:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.10.104)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1786385497'  
Notice: /Stage[main]/Dynamicinfra::Install/Package[tree]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Install/Package[apache2]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/ensure: defined content as '{sha256}b509069a1cd4daee1fd0a701cf6d3714fd4df70e0bfde0935e0580f76d6c74d8'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Scheduling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh' from 1 event  
Notice: /Stage[main]/Dynamicinfra::Report/File[/opt/company/deployment-report.txt]/ensure: defined content as '{sha256}53e42732a910eea5e1a52ef28d8b17ca66a9bc9c200e76d61493f6c20fb9156e'  
Notice: Applied catalog in 20.55 seconds  
ubuntu@puppet-agent01:~$ sudo cat /opt/company/system-info.txt  
=====
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo cat /opt/company/system-info.txt  
=====br/>ENTERPRISE INFRASTRUCTURE REPORTbr/>=====br/>  
Server Role      : Web Server  
Normalized Role  : web-server  
Environment      : Production  
Environment Label : PRODUCTION ENVIRONMENT  
Deployment Identity: PRODUCTION ENVIRONMENT -  
                  web-server  
  
Hostname         : puppet-agent01  
Primary IP       : 172.31.0.173  
  
Operating System : Ubuntu  
OS Family        : Debian  
  
Package Manager  : APT  
Package Command  : apt  
  
Generated By     : Vishnu  
=====br/>ubuntu@puppet-agent01:~$ sudo cat /opt/company/deployment-report.txt
```

```
ubuntu@puppet-agent01: ~  
Generated By      : Vishnu  
=====br/>ubuntu@puppet-agent01:~$ sudo cat /opt/company/deployment-report.txt  
=====br/>DEPLOYMENT REPORT  
=====br/>  
Server Role      : Web Server  
Environment      : Production  
  
Hostname         : puppet-agent01  
Primary IP       : 172.31.0.173  
Operating System : Ubuntu  
  
Status           : Managed by Vishnu  
=====br/>ubuntu@puppet-agent01:~$ ls -l /opt/company  
total 8  
-rw-r--r-- 1 root root 346 Aug 10 18:11 deployment-report.txt  
-rw-r--r-- 1 root root 561 Aug 10 18:11 system-info.txt  
ubuntu@puppet-agent01:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese  
   Active: active (running) since Mon 2026-08-10 18:11:58 UTC; 3min 56s ago
```

```
ubuntu@puppet-agent01: ~
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese>
   Active: active (running) since Mon 2026-08-10 18:11:58 UTC; 3min 56s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 2565 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU>
 Main PID: 2571 (apache2)
    Tasks: 55 (limit: 1018)
   Memory: 5.2M
      CPU: 38ms
   CGroup: /system.slice/apache2.service
           └─2571 /usr/sbin/apache2 -k start
             └─2572 /usr/sbin/apache2 -k start
               └─2573 /usr/sbin/apache2 -k start

Aug 10 18:11:58 puppet-agent01 systemd[1]: Starting The Apache HTTP Server...
Aug 10 18:11:58 puppet-agent01 systemd[1]: Started The Apache HTTP Server.
ubuntu@puppet-agent01:~$ dpkg -l | grep -E 'apache2|tree|curl|htop'
>
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-await/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                               Version
||/ Architecture Description
+++=====
```

```
ubuntu@puppet-agent01: ~
ii amd64 X authentication utility
ii xdg-user-dirs amd64 0.17-2ubuntu4
   tool to manage well known user directories
ii xfsprogs amd64 5.13.0-1ubuntu2.1
   Utilities for managing the XFS filesystem
ii xkb-data amd64 2.33-1
   all X Keyboard Extension (XKB) configuration data
ii xxd amd64 2:8.2.3995-1ubuntu2.34
   tool to make (or reverse) a hex dump
ii xz-utils amd64 5.2.5-2ubuntu1.1
   XZ-format compression utilities
ii zerofree amd64 1.1.1-1build3
   zero free blocks from ext2, ext3 and ext4 file-systems
ii zlib1g:amd64 amd64 1:1.2.11.dfsg-2ubuntu9.2
   compression library - runtime
ii zstd amd64 1.4.8+dfsg-3build1
   fast lossless compression algorithm -- CLI tool
ubuntu@puppet-agent01:~$ dpkg-query -W -f='${Package}: ${Status}\n' apache2 tree
curl htop
apache2: install ok installed
curl: install ok installed
htop: install ok installed
tree: install ok installed
ubuntu@puppet-agent01:~$
```